

Windows Domain Administration, Identity & Auditing

Seven support cases from my lab.local homelab

This is my first full Active Directory support project. I worked through seven small cases in my own VMs and kept the screenshots from each test. I tried to work the same way I would on a real ticket: check the current state, make one change, test it, and write down what actually happened. A few things did not behave the way I expected, so those are included too.

My lab

System / account	Role in the lab	Recorded state
DC01	Domain controller and DNS	Windows Server 2025 Standard Evaluation, 24H2, build 26100.32230. Lab IPv4: 192.168.169.10.
WIN11-IT-01	Domain workstation	Windows 11 Pro 25H2, build 26200.9457. Computer object in the Workstations OU.
LAB\ajaradi	Normal user	Used for ordinary client work.
LAB\adm-ajaradi	Separate admin account	Used for UAC prompts and elevated admin tools.
LAB\jonas	Helpdesk test account	Used to test the Finance-only password-reset delegation.
WIN11-IT-01\LocalAdmin	Local fallback account	Checked before the trust test so I still had a local way into the PC.

Cases in this project

ID	Test	Status
P1-T1	Health, DNS and domain trust	PASS
P1-T2	Standard vs administrative identity	PASS
P1-T3	Logon auditing with Group Policy	PASS
P1-T4	Scoped helpdesk password reset	PASS
P1-T5	GPMC support incident	PASS
Break/Fix 1	Broken secure channel and repair	PASS
Break/Fix 2	Protected AD object deletion	PASS

Task: Before I changed anything, I wanted to confirm the server/client versions, where the PC sits in AD, and whether the workstation still trusts the domain.

I started by checking the machines before changing anything. DC01 is the domain controller and WIN11-IT-01 is the client I use for the tests. I used winver as the final check for the Windows product and build numbers so the versions are clear in the evidence.

```
PS C:\Users\Administrator> hostname
whoami
Get-ComputerInfo | Select-Object WindowsProductName, WindowsVersion, OsBuildNumber
DC01
Tab\administrator

WindowsProductName      WindowsVersion  OsBuildNumber
-----
Windows Server 2025 Standard Evaluation 2009          26100

PS C:\Users\Administrator> |
```

E01 - DC01 hostname, administrative context and server build output.



E02 - winver confirms Windows Server 2025 Standard Evaluation, version 24H2, build 26100.32230.



E03 - winver confirms Windows 11 Pro 25H2, build 26200.9457.

```
PS C:\Users\Administrator> Get-ADComputer WIN11-IT-01 -Properties DistinguishedName |
Select-Object Name, DistinguishedName

Name      DistinguishedName
-----
WIN11-IT-01 CN=WIN11-IT-01,OU=Workstations,DC=lab,DC=local

PS C:\Users\Administrator>
```

E04 - WIN11-IT-01 is in OU=Workstations,DC=lab,DC=local.

PASS	Server/client versions and the Workstations OU placement are shown in the evidence.
------	---

The main health checks passed. WIN11-IT-01 could find DC01 and the secure channel was healthy. One thing stood out: DC01 had both 192.168.169.10 and a Tailscale address (100.x) in DNS. When I checked, the PC had actually found DC01 through the Tailscale address. I manage the lab over Tailscale, so I left the records alone during P1 and wrote it down to clean up later.

```
PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName lab.local -Name "*" -RRType A

HostName      RecordType Type      Timestamp      TimeToLive      RecordData
-----
e             A          1          25.09.2026 03:00:00 00:10:00      192.168.169.10
dc01          A          1          0              01:00:00
dc01          A          1          0              01:00:00      192.168.169.10
WIN11-IT-01   A          1          25.09.2026 03:00:00 00:20:00      192.168.169.130
WIN11-IT-02   A          1          12.09.2026 14:00:00 00:20:00      192.168.169.133

PS C:\Users\Administrator>
```

E05 - Current A records in lab.local, including DC01 and the client records.

```
PS C:\Users\Administrator> Resolve-DnsName DC01.lab.local

Name      Type      TTL      Section      IPAddress
----
DC01.lab.local AAAA      1200      Question
DC01.lab.local AAAA      1200      Question      fe80::109:28eb:ff91:9e68
DC01.lab.local A         1200      Question
DC01.lab.local A         1200      Question      192.168.169.10

PS C:\Users\Administrator> Resolve-DnsName DC01.lab.local

Name      Type      TTL      Section      IPAddress
----
DC01.lab.local AAAA      1200      Question
DC01.lab.local AAAA      1200      Question      fe80::109:28eb:ff91:9e68
DC01.lab.local A         1200      Question
DC01.lab.local A         1200      Question      192.168.169.10

PS C:\Users\Administrator> Get-NetIPAddress -IPAddress [redacted] | Select-Object InterfaceAlias, IPAddress

InterfaceAlias IPAddress
-----
Tailscale      [redacted]
```

E06 - Resolve-DnsName returns both the lab address and a Tailscale 100.x address; the adapter check confirms the 100.x address belongs to Tailscale.

```
PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName lab.local -Name DC01

HostName      RecordType Type      Timestamp      TimeToLive      RecordData
-----
DC01          A          1          0              01:00:00      192.168.169.10
DC01          AAAA      28          0              01:00:00

PS C:\Users\Administrator> Resolve-DnsName DC01.lab.local -DnsOnly

Name      Type      TTL      Section      IPAddress
----
DC01.lab.local AAAA      3600      Answer
DC01.lab.local A         3600      Answer      192.168.169.10
DC01.lab.local A         3600      Answer
```

E07 - The DC01 DNS record contains both 192.168.169.10 and a Tailscale 100.x address.

```
PS C:\Users\ajaradi> nltest /dsgetdc:lab.local
Test-ComputerSecureChannel -Verbose
DC: \DC01.lab.local
Address: \
Dns Guid: 6476acff-18bc0137dc6f
Dns Name: lab.local
Forest Name: lab.local
DC Site Name: Default-First-Site-Name
Our Site Name: Default-First-Site-Name
Flags: PDC GC DC LOW ADG TRUSTED GTMSESRV WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE FULL_SECRET WS DS_8 DS_9 DS_10 KEYLIST DS_13
The command completed successfully
Verbose: Performing the operation "Test-ComputerSecureChannel" on target "WIN11-IT-01".
True
Verbose: The secure channel between the local computer and the domain lab.local is in good condition.

PS C:\Users\ajaradi>
```

E08 - WIN11-IT-01 locates DC01.lab.local through the Tailscale 100.x address, and Test-ComputerSecureChannel reports a healthy relationship.

PASS

WIN11-IT-01 can find DC01 and the secure channel is healthy. The extra Tailscale DNS record is noted for later.

Task: Use a normal account for daily work and a separate account only when I actually need admin rights.

My normal desktop runs as LAB\ajaradi. The captured token is Medium integrity, and ajaradi is not in the local Administrators group. When I choose Run as administrator, Windows asks for another account. I use LAB\adm-ajaradi for that, and only the new PowerShell window runs elevated at High integrity.

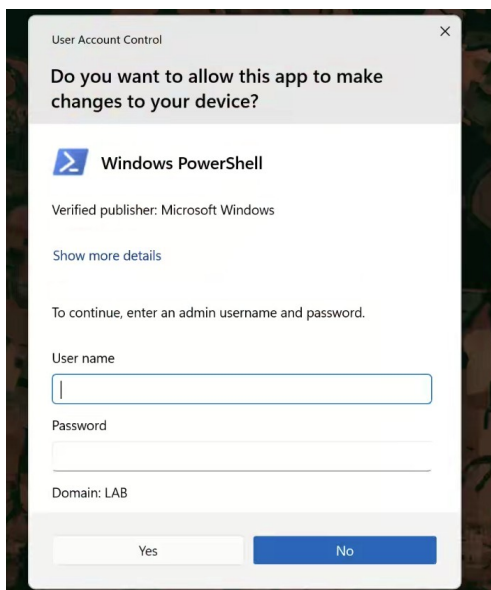
```
PS C:\Users\ajaradi> whoami
whoami /groups
net localgroup Administrators
lab\ajaradi

GROUP INFORMATION
-----
Group Name                                     Type                SID                  Attributes
-----
Everyone                                     Well-known group    S-1-1-0             Mandatory group, Enabled by default, Enabled group
BUILTIN\Users                               Alias               S-1-5-32-545        Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                     Well-known group    S-1-5-4             Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON                               Well-known group    S-1-2-1             Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users             Well-known group    S-1-5-11            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization                Well-known group    S-1-5-15            Mandatory group, Enabled by default, Enabled group
LOCAL                                         Well-known group    S-1-2-0             Mandatory group, Enabled by default, Enabled group
Authentication authority asserted identity    Well-known group    S-1-18-1            Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Mandatory Level       Label               S-1-16-8192
Alias name                                     Administrators
Comment                                     Administrators have complete and unrestricted access to the computer/domain

Members
-----
Administrator
LAB\Domain Admins
LocalAdmin
The command completed successfully.

PS C:\Users\ajaradi>
```

E09 - Normal LAB\ajaradi shell: Medium integrity; ajaradi is not listed in the local Administrators group.



E10 - UAC asks the standard user for administrator credentials.

```
PS C:\WINDOWS\system32> whoami
whoami /groups | Findstr /i "Administrators Domain Mandatory"
lab\adm-ajaradi
Everyone                                     Well-known group    S-1-1-0             Mandatory group, Enabled by default, Enabled group
BUILTIN\Users                               Alias               S-1-5-32-545        Mandatory group, Enabled by default, Enabled group
BUILTIN\Administrators                     Alias               S-1-5-32-544        Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                     Well-known group    S-1-5-4             Mandatory group, Enabled by default, Enabled group
CONSOLE LOGON                               Well-known group    S-1-2-1             Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users             Well-known group    S-1-5-11            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization                Well-known group    S-1-5-15            Mandatory group, Enabled by default, Enabled group
LOCAL                                         Well-known group    S-1-2-0             Mandatory group, Enabled by default, Enabled group
LAB\Domain Admins                           Group               S-1-5-22-3735704636-2243183458-1193105392-512 Mandatory group, Enabled by default, Enabled group
Authentication authority asserted identity    Well-known group    S-1-18-1            Mandatory group, Enabled by default, Enabled group
LAB\Limited RODC Password Replication Group Alias               S-1-5-22-3735704636-2243183458-1193105392-572 Mandatory group, Enabled by default, Enabled group
Mandatory Label\High Mandatory Level         Label               S-1-16-12288
PS C:\WINDOWS\system32>
```

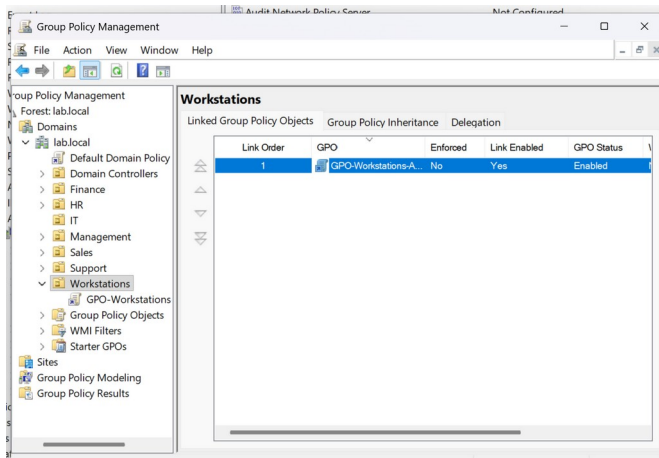
E11 - Elevated PowerShell runs as LAB\adm-ajaradi with High integrity and administrative group membership.

PASS

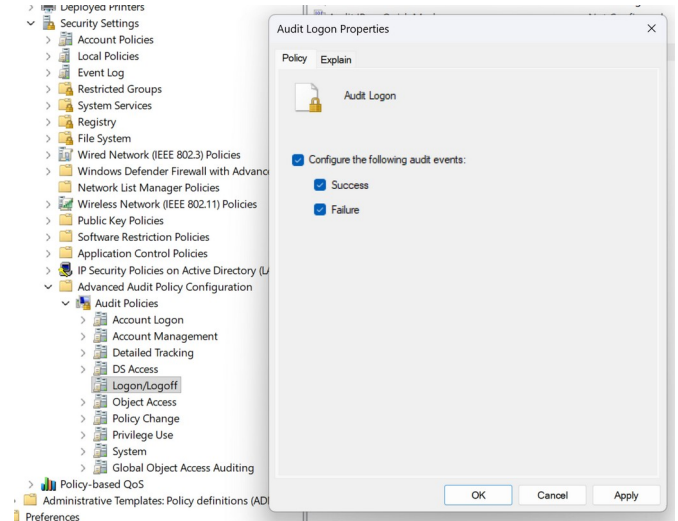
ajaradi stays a standard user. Admin work runs in a separate elevated process as adm-ajaradi.

Task: Record both successful and failed logons on WIN11-IT-01 and make sure the GPO actually reaches the client.

I created the audit GPO, linked it to the Workstations OU, refreshed policy on WIN11-IT-01, and checked auditpol. I wanted to see the Success and Failure setting on the PC itself, not only in the GPO console.



E12 - GPO-Workstations-Audit-Logon linked to the Workstations OU.



E13 - Audit Logon configured for Success and Failure.

```
PS C:\WINDOWS\system32> whoami
lab\adm-ajaradi
PS C:\WINDOWS\system32> gpupdate /target:computer /force
Updating policy...

Computer Policy update has completed successfully.
PS C:\WINDOWS\system32>
```

E14 - Admin context recorded; computer policy refresh completes successfully.

```
PS C:\WINDOWS\system32> auditpol /get /subcategory:"Logon"
System audit policy
Category/Subcategory      Setting
Logon/Logoff
Logon                      Success and Failure
PS C:\WINDOWS\system32>
```

E15 - Effective local audit setting reports Logon = Success and Failure.

PASS

The GPO is configured and the client shows Logon auditing set to Success and Failure.

Next I used gpresult to make sure the GPO actually reached WIN11-IT-01. Then I used runas twice: first with a wrong password and then with the right one. That gave me one failed and one successful logon to look for in the Security log.

```
PS C:\WINDOWS\system32> gpresult /scope computer /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 9/22/2026 at 12:26:41 AM

RSOP data for on WIN11-IT-01 : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:             Default-First-Site-Name
Roaming Profile:
Local Profile:
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=WIN11-IT-01,OU=Workstations,DC=lab,DC=local
Last time Group Policy was applied: 9/22/2026 at 12:22:59 AM
Group Policy was applied from: DC01.lab.local
Group Policy slow link threshold: 500 kbps
Domain Name: LAB
Domain Type: Windows 2008 or later

Applied Group Policy Objects
-----
GPO-Workstations-Audit-Logon
Default Domain Policy

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The computer is a part of the following security groups
-----
BUILTIN\Administrators
Everyone
BUILTIN\Users
NT AUTHORITY\NETWORK
NT AUTHORITY\Authenticated Users
This Organization
WIN11-IT-01$
Domain Computers
Authentication authority asserted identity
System Mandatory Level

PS C:\WINDOWS\system32>
```

E16 - gpresult shows WIN11-IT-01 in the Workstations OU with the audit GPO applied.

```
C:\Users\ajardi>runas /user:LAB\ajardi cmd
Enter the password for LAB\ajardi:
Attempting to start cmd as user "LAB\ajardi" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\ajardi>
```

E17 - Deliberately incorrect runas password; error 1326.

```
C:\Users\ajardi>runas /user:LAB\ajardi cmd
Enter the password for LAB\ajardi:
Attempting to start cmd as user "LAB\ajardi" ...

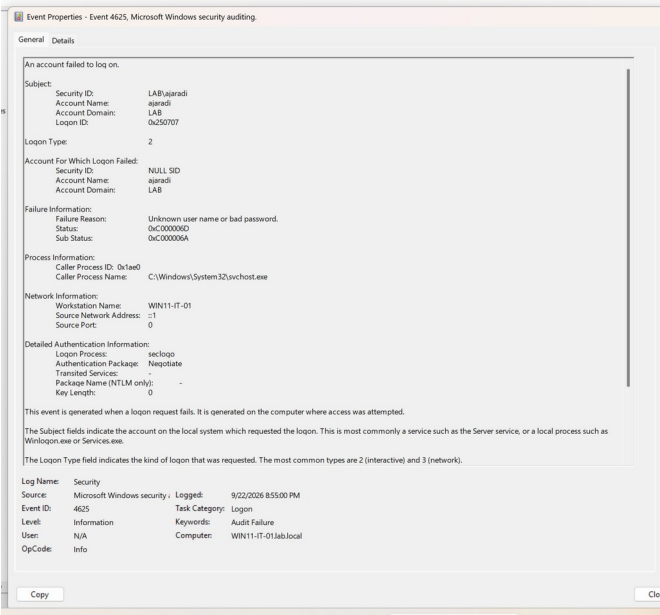
C:\Users\ajardi>
```

cmd (running as LAB\ajardi) x +

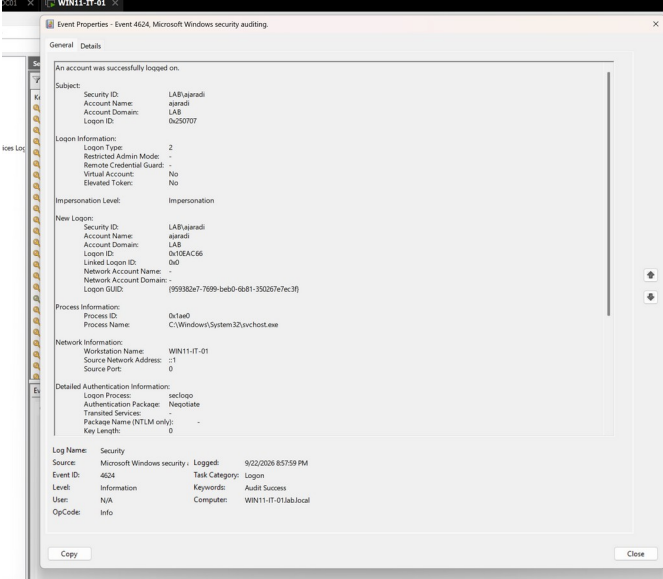
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Windows\System32>

E18 - Correct credentials open a command window as LAB\ajardi.

Event Viewer showed both results. The failed attempt is Event 4625, Logon Type 2, for LAB\ajaradi with the bad-password status values. The successful attempt is Event 4624, also Type 2, and the captured event shows Elevated Token = No.



E19 - Event 4625: failed Type 2 logon for LAB\ajaradi; status 0xC000006D, substatus 0xC000006A.



E20 - Event 4624: successful Type 2 logon for LAB\ajaradi; Elevated Token = No.

PASS

The policy is applied and the Security log contains both the failed and successful test logons.

Task: Let Jonas reset Finance passwords, but not HR passwords, and do it without making him an administrator.

I kept this inside PortfolioLab and used throwaway Finance and HR users. The permission went to the GG_PortfolioLab_Helpdesk group, not straight to Jonas. In the delegation wizard I gave that group only the password-reset task on the Finance OU.

```
PS C:\Users\Administrator> New-ADOrganizationalUnit -Name "PortfolioLab" -Path "DC=lab,DC=local"
"Finance","HR","Disabled" | ForEach-Object {
    New-ADOrganizationalUnit -Name $_ -Path "OU=PortfolioLab,DC=lab,DC=local"
}
Get-ADOrganizationalUnit -Filter * -SearchBase "OU=PortfolioLab,DC=lab,DC=local" |
Select-Object Name, DistinguishedName
```

Name	DistinguishedName
PortfolioLab	OU=PortfolioLab,DC=lab,DC=local
Finance	OU=Finance,OU=PortfolioLab,DC=lab,DC=local
HR	OU=HR,OU=PortfolioLab,DC=lab,DC=local
Disabled	OU=Disabled,OU=PortfolioLab,DC=lab,DC=local

```
PS C:\Users\Administrator>
```

E21 - PortfolioLab with Finance, HR and Disabled child OUs.

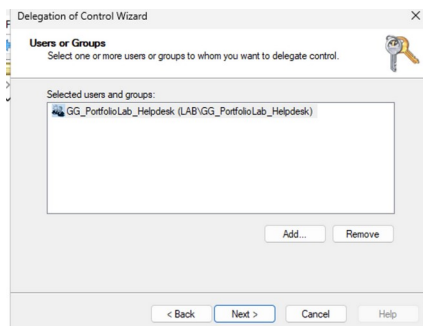
```
PS C:\WINDOWS\system32> Get-ADUser -Filter * `
-SearchBase "OU=PortfolioLab,DC=lab,DC=local" |
Select-Object Name, SamAccountName, DistinguishedName

Get-ADGroupMember "GG_PortfolioLab_Helpdesk" |
Select-Object Name, SamAccountName
```

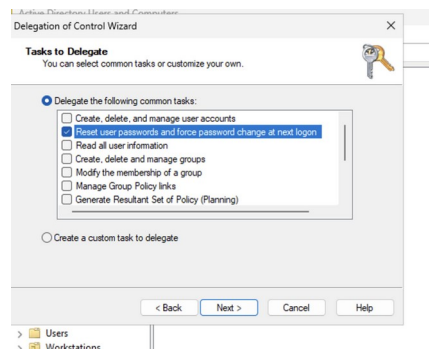
Name	SamAccountName	DistinguishedName
Finance Test01	finance.test01	CN=Finance Test01,OU=Finance,OU=PortfolioLab,DC=lab,DC=local
HR Test01	hr.test01	CN=HR Test01,OU=HR,OU=PortfolioLab,DC=lab,DC=local
Jonas Helpdesk	jonas	CN=Jonas Helpdesk,OU=PortfolioLab,DC=lab,DC=local

```
PS C:\WINDOWS\system32>
```

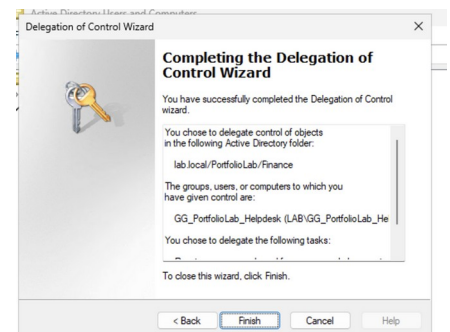
E22 - Finance Test01, HR Test01, Jonas Helpdesk and the helpdesk security group verified.



E23 - GG_PortfolioLab_Helpdesk selected in Delegation of Control.



E24 - Only Reset user passwords and force password change at next logon is selected.



E25 - Wizard summary shows the Finance OU and the delegated helpdesk group.

I checked the Finance OU ACL, then tested the permission as LAB\jonas. I made one mistake first: I was still in an Administrator window, so that result proved nothing. I threw that test away and opened a PowerShell window as Jonas. From there the Finance reset worked, the HR reset was denied, and Jonas did not show Domain Admins or BUILTIN\Administrators in the captured group check.

```
PS C:\WINDOWS\system32> dsac ls "OU=Finance,OU=PortfolioLab,DC=lab,DC=local" | findstr /i "Helpdesk"
dsac ls "OU=HR,OU=PortfolioLab,DC=lab,DC=local" | findstr /i "Helpdesk"
Allow LAB\GG_PortfolioLab_Helpdesk Reset Password
Allow LAB\GG_PortfolioLab_Helpdesk SPECIAL ACCESS for pwdLastSet

PS C:\WINDOWS\system32>
```

E26 - dsac ls shows Reset Password and related pwdLastSet rights for the helpdesk group on the Finance OU.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> runas /user:LAB\jonas powershell.exe
Enter the password for LAB\jonas:
Attempting to start powershell.exe as user "LAB\jonas" ...
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> powershell.exe (running as LAB\jonas)
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> whoami
LAB\jonas
PS C:\WINDOWS\system32>
```

E27 - Test shell launched as LAB\jonas on WIN11-IT-01.

```
PS C:\WINDOWS\system32> whoami
LAB\jonas
PS C:\WINDOWS\system32> $new = Read-Host "New password for Finance Test01" -AsSecureString
>>
>> try {
>>     Set-ADAccountPassword `
>>     -Identity "finance.test01" `
>>     -Reset `
>>     -NewPassword $new `
>>     -ErrorAction Stop
>> }
>> "RESULT: FINANCE PASSWORD RESET SUCCESS"
>> }
>> catch {
>>     "RESULT: FINANCE PASSWORD RESET FAILED"
>>     $_.Exception.Message
>> }
New password for Finance Test01: *****
RESULT: FINANCE PASSWORD RESET SUCCESS
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
```

E29 - As Jonas, Finance Test01 password reset succeeds.

```
PS C:\WINDOWS\system32> whoami
LAB\jonas
PS C:\WINDOWS\system32> whoami /groups | Select-String "Domain Admins|BUILTIN\Administrators"
LAB\jonas
PS C:\WINDOWS\system32>
```

E28 - Captured Jonas token does not show Domain Admins or BUILTIN\Administrators.

```
PS C:\WINDOWS\system32> try {
>>     Set-ADAccountPassword `
>>     -Identity "hr.test01" `
>>     -Reset `
>>     -NewPassword $new `
>>     -ErrorAction Stop
>> }
>> "RESULT: HR PASSWORD RESET SUCCESS"
>> }
>> catch {
>>     "RESULT: HR PASSWORD RESET DENIED"
>>     $_.Exception.Message
>> }
RESULT: HR PASSWORD RESET DENIED
Access is denied
PS C:\WINDOWS\system32>
```

E30 - The same Jonas process receives Access is denied when resetting HR Test01.

PASS

Jonas can reset Finance Test01. The same reset against HR Test01 is denied.

Task: gpmc.msc does not open on WIN11-IT-01. Fix the admin tool and make sure it works afterward.

gpmc.msc was not recognised on WIN11-IT-01. I installed the RSAT Group Policy Management capability and tried again. GPMC opened normally after that. I didn't check first whether the tool was installed, so I can't prove it was missing, only that installing it fixed the problem.

```
PS C:\WINDOWS\system32> gpmc.msc
gpmc.msc : The term 'gpmc.msc' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ gpmc.msc
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (gpmc.msc:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\system32>
```

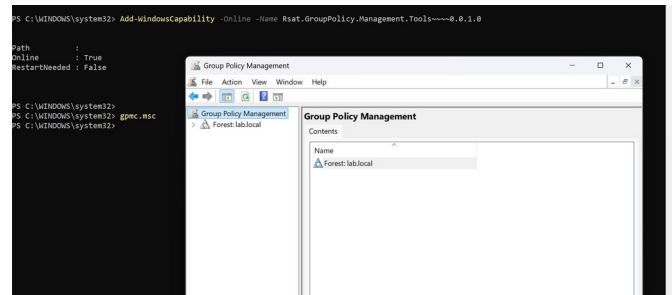
E31 - Initial symptom: gpmc.msc is not recognised on WIN11-IT-01.

```
PS C:\WINDOWS\system32> Add-WindowsCapability -Online -Name Rsat.GroupPolicy.Management.Tools~~~~0.0.1.0

Path      :
Online    : True
RestartNeeded : False

PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
```

*E32 - RSAT Group Policy Management capability installation completes;
RestartNeeded = False.*



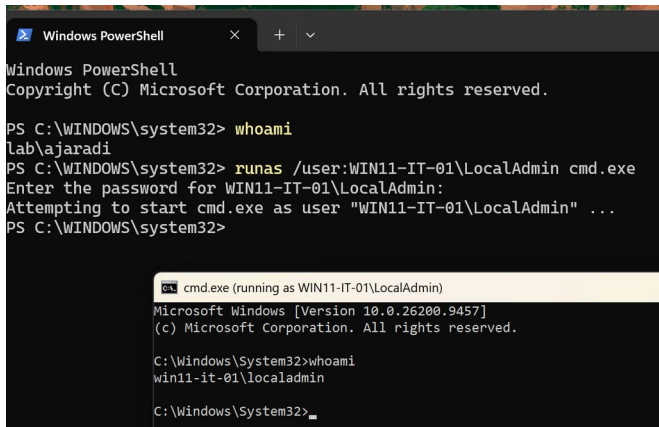
E33 - Retest: Group Policy Management opens and shows Forest: lab.local.

PASS

GPMC opens after the RSAT Group Policy Management capability is installed.

Task: Reproduce a broken workstation trust in the lab, keep LocalAdmin available as a fallback, then repair the secure channel without a domain rejoin.

Before I broke anything, I checked the local LocalAdmin account and made sure it worked without the domain. I also captured the secure channel while it was still healthy. That gave me a way back into the client if the domain login stopped working.



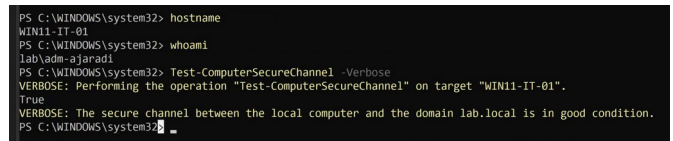
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> whoami
lab\ajaradi
PS C:\WINDOWS\system32> runas /user:WIN11-IT-01\LocalAdmin cmd.exe
Enter the password for WIN11-IT-01\LocalAdmin:
Attempting to start cmd.exe as user "WIN11-IT-01\LocalAdmin" ...
PS C:\WINDOWS\system32>

cmd.exe (running as WIN11-IT-01\LocalAdmin)
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

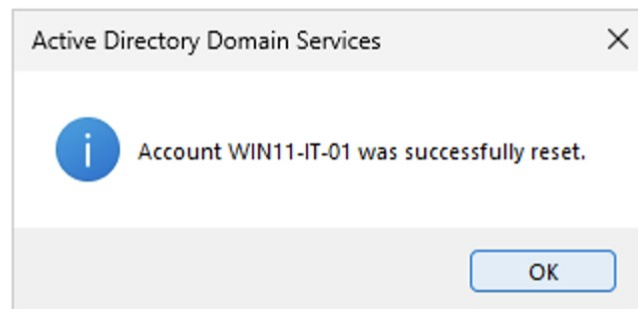
C:\Windows\System32>whoami
win11-it-01\localadmin
C:\Windows\System32>
```

E34 - Local recovery account verified before the test.



```
PS C:\WINDOWS\system32> hostname
WIN11-IT-01
PS C:\WINDOWS\system32> whoami
lab\adm-ajaradi
PS C:\WINDOWS\system32> Test-ComputerSecureChannel -Verbose
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "WIN11-IT-01".
True
VERBOSE: The secure channel between the local computer and the domain lab.local is in good condition.
PS C:\WINDOWS\system32>
```

E35 - Baseline: WIN11-IT-01 secure channel returns True and reports good condition.



E36 - AD DS confirms that the WIN11-IT-01 computer account was reset.

This one took a while. Right after the computer-account reset, the open client session still reported a healthy channel. I did not count that as a failure. After I restarted WIN11-IT-01, the same test returned False and reported that the secure channel was broken.

After the restart, the client finally showed the trust problem: Test-ComputerSecureChannel returned False. I repaired it from the client with Test-ComputerSecureChannel -Repair and my separate admin credentials. Then I ran the test again instead of assuming the repair worked.

```
PS C:\WINDOWS\system32> hostname
whoami
Test-ComputerSecureChannel -Server DC01.lab.local -Verbose
WIN11-IT-01
lab\adm-ajaradi
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "WIN11-IT-01".
False
VERBOSE: The secure channel between the local computer and the domain lab.local is broken.

PS C:\WINDOWS\system32> |
```

E37 - Broken state: Test-ComputerSecureChannel returns False and reports the channel is broken.

```
$cred = Get-Credential "LAB\adm-ajaradi"
Test-ComputerSecureChannel -Repair -Server DC01.lab.local -Credential $cred -Verbose
```

```
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> $cred = Get-Credential "LAB\adm-ajaradi"
Test-ComputerSecureChannel -Repair -Server DC01.lab.local -Credential $cred -Verbose
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "WIN11-IT-01".
True
VERBOSE: The secure channel between the local computer and the domain lab.local was successfully repaired.
PS C:\WINDOWS\system32>
```

E38 - Repair command returns True and reports that the secure channel was successfully repaired.

```
VERBOSE: The secure channel between the local computer and the domain lab.local was successfully repaired.
PS C:\WINDOWS\system32> Test-ComputerSecureChannel -Server DC01.lab.local -Verbose
VERBOSE: Performing the operation "Test-ComputerSecureChannel" on target "WIN11-IT-01".
True
VERBOSE: The secure channel between the local computer and the domain lab.local is in good condition.
PS C:\WINDOWS\system32>
```

E39 - Independent retest returns True and reports the secure channel is in good condition.

PASS

The secure channel is healthy again, and WIN11-IT-01 never had to leave lab.local.

Task: A disposable AD user returns Access is denied even from an admin session. Check the object before changing permissions.

The delete returned Access is denied even though I was using an admin account. Instead of changing permissions, I checked the object itself. ProtectedFromAccidentalDeletion was True. I turned that protection off, deleted the test user, and confirmed with Get-ADUser that the object was gone.

```
PS C:\Users\Administrator> whoami
New-ADUser -Name "Delete Test" -SamAccountName "delete.test01" -Path "OU=Disabled,OU=PortfolioLab,DC=lab,DC=local" -Enabled $false
Set-ADObject -Identity "CN=Delete Test,OU=Disabled,OU=PortfolioLab,DC=lab,DC=local" -ProtectedFromAccidentalDeletion $true
Remove-ADUser -Identity delete.test01 -Confirm:$false
Tab\administrator
Remove-ADUser : Access is denied
At line:4 char:1
+ Remove-ADUser -Identity delete.test01 -Confirm:$false
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (delete.test01:ADUser) [Remove-ADUser], UnauthorizedAccessException
+ FullyQualifiedErrorId : ActiveDirectoryCmdlet:System.UnauthorizedAccessException,Microsoft.ActiveDirectory.Management.Commands.RemoveADUser

PS C:\Users\Administrator>
```

E40 - Initial delete attempt returns Access is denied.

```
PS C:\Users\Administrator> Get-ADObject `
-Identity "CN=Delete Test,OU=Disabled,OU=PortfolioLab,DC=lab,DC=local" `
-Properties ProtectedFromAccidentalDeletion |
Select-Object Name, ProtectedFromAccidentalDeletion

Name      ProtectedFromAccidentalDeletion
-----
Delete Test      True

PS C:\Users\Administrator>
```

E41 - ProtectedFromAccidentalDeletion is explicitly confirmed as True.

```
PS C:\Users\Administrator> Set-ADObject `
-Identity "CN=Delete Test,OU=Disabled,OU=PortfolioLab,DC=lab,DC=local" `
-ProtectedFromAccidentalDeletion $false
Remove-ADUser -Identity delete.test01 -Confirm:$false
Get-ADUser -Identity delete.test01
Get-ADUser : Cannot find an object with identity: 'delete.test01' under: 'DC=lab,DC=local'.
At line:10 char:1
+ Get-ADUser -Identity delete.test01
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (delete.test01:ADUser) [Get-ADUser], ADIdentityNotFoundException
+ FullyQualifiedErrorId : ActiveDirectoryCmdlet:Microsoft.ActiveDirectory.Management.ADIdentityNotFoundException,Microsoft.ActiveDirectory.Management.Commands.GetADUser

PS C:\Users\Administrator>
```

E42 - After removing protection and deleting the user, Get-ADUser reports the object cannot be found.

PASS

Accidental-deletion protection was on. After I turned it off, the delete worked and the user could no longer be found.

The part I found most useful was having to prove each change. I had to check whether the GPO really reached the PC, whether Jonas could do only the job I gave him, and whether the broken trust was actually repaired. The trust test took the longest because the failure did not show until I restarted the client.

What I practised

- Active Directory computer and user administration in lab.local
- DNS record inspection and domain-controller discovery
- Separate standard and administrative identities with UAC
- Group Policy configuration plus gpupdate, auditpol and gpresult verification
- Windows Security log review for Events 4624 and 4625
- Delegation of Control with a positive test and an out-of-scope negative test
- RSAT troubleshooting on a Windows 11 administrative client
- Secure-channel diagnosis and repair without a domain rejoin
- ProtectedFromAccidentalDeletion troubleshooting on a disposable AD object

Limitations

- This lab has one domain controller, so I did not test AD replication or redundancy.
- lab.local is only for this lab. In a real environment I would use a namespace based on a domain the organisation owns.
- Windows Server is an evaluation installation.
- DC01 still has both 192.168.169.10 and a Tailscale address (100.x) registered in DNS. I left that alone during P1 and wrote it down for later.
- For the GPMC case, I captured the error, the RSAT install and the working console, but not the feature state before I installed it.

Next

P2 will add a file server and move into Finance and HR share permissions, role changes and file recovery. I will keep the same domain so the projects build on each other.

Evidence note: All screenshots are from this lab. Passwords are not shown. E01-E42 are just labels so I can find the original screenshots later.